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First detection of West Nile virus in UK mosquitoes

UK findings

The Animal and Plant Health Agency (APHA) and UK Health Security Agency (UKHSA) have been conducting research (https://www.vb-radar.com/) looking for evidence of mosquito borne viruses entering the country by testing birds and mosquitoes. In May 2025, APHA reported that fragments of West Nile virus (WNV) genome had been detected by PCR testing in 2 pools of Aedes vexans mosquitoes (each of 10 mosquitoes) collected by UKHSA in Nottinghamshire during 2023. This is the first evidence of WNV in mosquitoes in the UK. A total of 200 pools of mosquitoes from the same site were PCR tested, with 198 pools testing negative. A further 30,000 mosquitoes and 300 birds have been PCR tested from other areas in England most suitable for WNV, with all samples from 2023 and 2024 testing negative.

This isolated finding was limited to one small site in Gamston (Retford), Nottinghamshire, England, and there is currently no evidence of ongoing epizootic activity anywhere in the UK. Aedes vexans are native to the UK and although they can be found widely at very low densities, in a few locations in England they can occur at high densities associated with summer flooded river landscapes. While uncommon, they are known to cause nuisance biting in a small number of areas in the country, including in parts of Nottinghamshire. Land use changes in the affected area have significantly reduced mosquito numbers.

While Aedes vexans is potentially a competent vector for WNV (in other words, able to transmit virus between animals), Culex modestus is still considered the primary bridge vector (see below) for WNV and has not been found as far north as Nottinghamshire. All Culex modestus mosquitoes tested to date in England were negative for WNV.

The risk of a human outbreak is considered to be very low (https://www.gov.uk/guidance/west-nile-virus#risk-to-the-uk). To date, no locally acquired human cases of WNV have been reported in the UK, although a small number of travel-associated cases have occurred in the past. The main risk of WNV for UK residents continues to be travel to endemic areas overseas.

Background

WNV is a vector borne disease belonging to the Flaviviridae family, which also includes other viruses such as dengue (https://www.gov.uk/government/collections/dengue-fever-guidance-data-and-analysis) and yellow fever (https://www.gov.uk/guidance/yellow-fever-guidance-data-and-analysis). The natural reservoir of WNV is birds, and the virus typically circulates between birds and mosquitoes through bird-biting mosquitoes, particularly Culex pipiens. Other mosquitoes may bite both birds and humans (known as bridge vectors) and have the potential to transmit WNV to humans, with Culex modestus mosquitoes implicated as the principal vector in outbreaks of WNV in humans and horses in Europe. Other potential bridge vectors may include Aedes vexans mosquitoes.

Most people infected with WNV are asymptomatic or present with mild symptoms such as influenza-like illness (fever, headache or myalgia), swollen lymph nodes and a rash. While most mild infections resolve within 3 to 6 days, a small number of cases (under 1%) are severe and may experience neuroinvasive disease, typically presenting as meningitis, encephalitis or acute flaccid paralysis (sudden weakness), with high fever, neck stiffness, disorientation/confusion, severe muscle weakness, tremors, convulsions, paralysis and coma. Other neurological symptoms may include unsteadiness, weakness of facial muscles, visual disturbance or eye pain (see WNV symptoms (https://www.gov.uk/guidance/west-nile-virus#symptoms)).

Diagnosis is made by sending a blood sample, in the first instance, to the UKHSA Rare and Imported Pathogen's Laboratory (https://www.gov.uk/government/collections/rare-and-imported-pathogens-laboratory-ripl) (RIPL). Cases of unexplained encephalitis or suspected WNV encephalitis should be discussed at multi-disciplinary encephalitis meetings, including infectious diseases, microbiology and virology doctors, neurologists and specialist radiologists (see WNV diagnosis (https://www.gov.uk/guidance/west-nile-virus#diagnosis)). There is no available treatment, and no human vaccination.

The first large outbreak of WNV in Europe occurred in Romania in 1996 and, since then, human cases have been identified in several European countries including France, Italy, Portugal, Spain, Poland and Germany.

Factors including climate change and changes in land use have likely increased the geographical distribution of vectors capable of spreading diseases such as WNV, which may be introduced to a non-endemic country when an infected, migratory bird arrives having been bitten by a mosquito in an endemic area.

Further information

Key points and sources of further information are as follows:

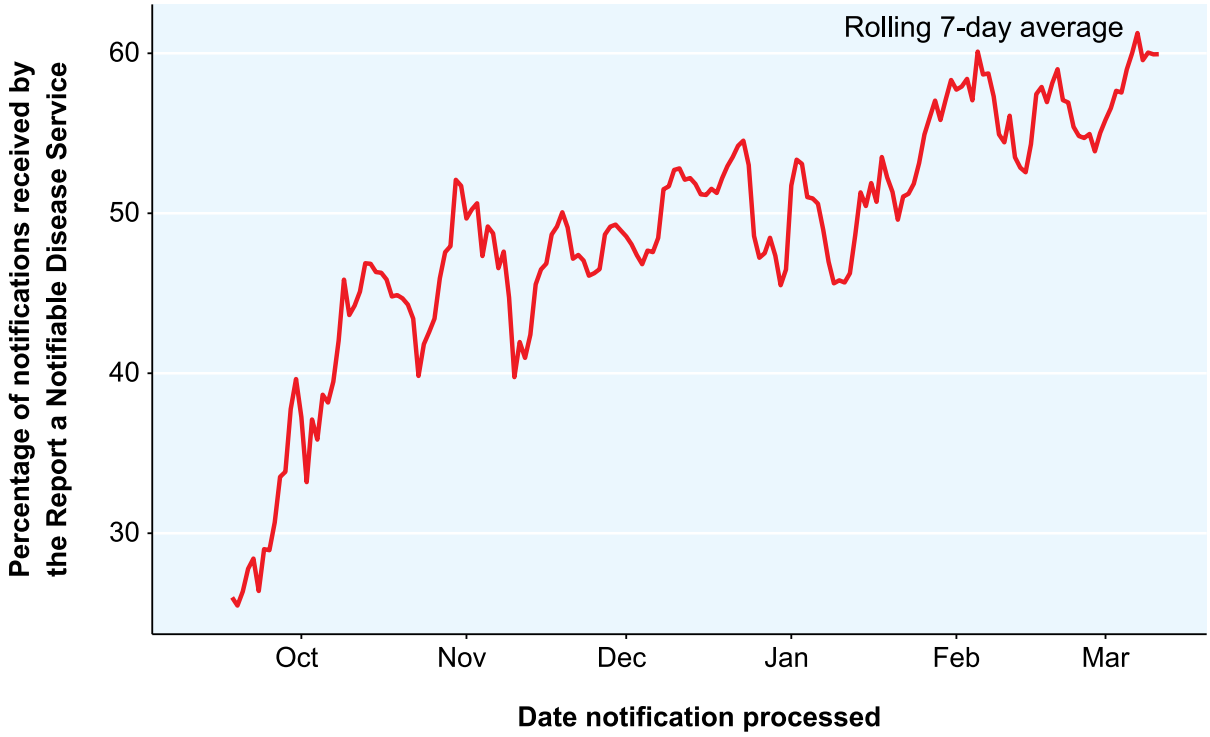
- recently updated clinical advice on West Nile virus including symptoms, diagnosis and epidemiology, is available on the UKHSA guidance page at: West Nile virus: epidemiology, diagnosis and prevention (https://www.gov.uk/guidance/west-nile-virus)
- West Nile virus is a notifiable organism and should be reported (https://www.gov.uk/guidance/notifiable-organisms-and-how-to-report-them) to UKHSA if detected
- acute encephalitis (including suspected WNV encephalitis) is a notifiable disease and should be reported (https://www.gov.uk/guidance/notifiable-diseases-and-how-to-report-them) to UKHSA.
- the European Centre for Disease Prevention and Control (ECDC) publishes surveillance data and updates for West Nile virus infection (https://www.ecdc.europa.eu/en/infectious-disease-topics/west-nile-virus-infection/surveillance-and-updates-west-nile-virus).

Successful introduction of a new online service for notifiable disease reporting

In September 2024, UKHSA launched 'Report a notifiable disease (https://www.gov.uk/guidance/report-a-notifiable-disease)' on GOV.UK. This is a new online service to assist Registered Medical Practitioners (RMPs) report legally notifiable diseases more quickly and easily to the Agency. Principal benefits of this development are faster public health action to reduce disease transmission and support prompt response to infectious disease outbreaks, and improved local and national surveillance.

Since the launch of this new service 6 months ago, UKHSA has seen a rise in electronic notifications and in March 2025 received 64% of all notifications electronically via the tool (see figure).

Percentage of all notifiable diseases reported online since September 2024



In the last 6 months UKHSA has received 10,406 notifications via the new service. Scarlet fever, food poisoning and whooping cough notifications formed over three quarters of all notifications; the chart below shows electronically reported notifications broken down by disease category.

Electronically made notifications, by disease category

Change to table view

Disease category	Notifications (%)
Scarlet fever	3,484 (33)
Food poisoning	2,498 (25)
Whooping cough	1,685 (16)
Mumps	870 (8)
Measles	517 (5)
All other notifiable diseases	1,352 (13)

Following analysis of the notification data, a few issues were identified leading to changes to the new arrangements. RMPs are now able to: subcategorise notifications; indicate whether a person has been vaccinated when a vaccine-preventable disease is being reported; and add further details about the case in a free text field.

Registered Medical Practitioners in the UK have a legal duty to notify suspected or confirmed cases of diseases prescribed under the Health Protection (Notification) Regulations 2010 (https://www.legislation.gov.uk/uksi/2010/659/contents/made). On 6 April 2025, these Regulations were updated and 8 new diseases became notifiable (see 'Changes to health protection notification regulations (https://ukhsa.blog.gov.uk/2025/03/11/changes-to-health-protection-notification-regulations/)').

UKHSA would like to encourage all registered medical practitioners to use 'Report a notifiable disease (https://www.gov.uk/guidance/report-a-notifiable-disease)' to streamline the notification process of non-urgent infectious diseases. All urgent infectious disease reports should be telephoned to the relevant local UKHSA Health Protection Team (https://www.gov.uk/health-protection-team).

Management of contacts of iGAS in community settings: a reminder

Due to the severity of invasive Group A streptococcal (iGAS) infections, and the immediate risk to close contacts, early recognition and prompt notification of cases is critical to ensure timely public health actions are taken to prevent further cases.

UKHSA published revised case definitions and public health guidelines for the management of contacts of iGAS in community settings in December 2022, replacing national guidelines published in 2004.

Frontline clinicians are reminded of the following three key changes introduced by the revised guidance (https://www.gov.uk/government/publications/invasive-group-a-streptococcal-disease-managing-community-contacts):

- expansion of case definitions
- a requirement for prompt notification of clinically suspected (probable) cases, even in the absence of microbiological confirmation
- the updating of the risk groups who should receive antibiotic prophylaxis

Expansion of case definitions

The definition of a Confirmed Case was expanded in line with healthcare and maternity GAS guidelines (2012); the definition now includes:

- GAS isolated from normally non-sterile sites such as throat, sputum, vagina or wound in combination with a severe clinical presentation (such as Streptococcal Toxic Shock Syndrome (STSS), necrotising fasciitis, pneumonia, septic arthritis, meningitis, peritonitis, osteomyelitis, myositis and puerperal sepsis) **together with**
- GAS isolated from a normally sterile site

A Probable Case definition was also included: that is, an individual with a severe clinical presentation consistent with iGAS infection – such as STSS, necrotising fasciitis, myositis, and puerperal sepsis – in the absence of microbiological confirmation of GAS and:

- the clinician considers that GAS is the most likely cause

or:

- there is an epidemiological link to a confirmed GAS case

Notification of confirmed and probable cases triggering public health action

To facilitate initiation of urgent public health actions, clinicians are now requested to notify their local health protection team (HPT) (https://www.gov.uk/health-protection-team), within 24 hours, of all probable cases on the basis of clinical suspicion (see case definitions above); notification should not be dependent on microbiological confirmation.

Information on how to notify can be found here: https://www.gov.uk/guidance/notifiable-diseases-and-causative-organisms-how-to-report (https://www.gov.uk/guidance/notifiable-diseases-and-causative-organisms-how-to-report)

All iGAS isolates from sterile sites should be sent to the UKHSA Staphylococcus and Streptococcus Reference Section (for England, Wales and Northern Ireland) or to the Scottish Microbiology Reference Laboratory (for Scotland).

Updated risk groups for antibiotic prophylaxis

A review of evidence led to the updating of recommendations for antibiotic prophylaxis for close contacts; this now extends beyond those who are symptomatic, and to mothers and babies where either develops iGAS in the post-partum period (as per the previous guidance), so as to also include:

- pregnant women from ≥37 weeks gestation
- neonates and women within the first 28 days of delivery
- older household contacts (≥75 years)
- individuals with active chickenpox lesions

The local HPT will facilitate the following actions for close contacts as follows:

Where individuals in risk groups are identified as a close contact of an iGAS case (confirmed or probable) in the 7 days prior to diagnosis – or within 24 hours of the case commencing antibiotics. Such individuals should be routinely offered antibiotic prophylaxis, if within 10 days of iGAS diagnosis in the index case.

Where close contacts have symptoms suggestive of localised GAS infection (sore throat, skin infection, fever) within 30 days of diagnosis in the index case, they should be assessed by a clinician and antibiotic treatment offered if GAS infection is suspected.

All close contacts should be provided with information on signs and symptoms and action to take should they develop symptoms with 30 days of diagnosis of the index case.

Evidence base for the revision of the guidance

The evidence for these changes resulted from expert opinion provided by the UK iGAS Community Guidelines Working Group and an extensive literature review (https://doi.org/10.1016/j.jinf.2025.106468).

Infection reports in this issue

Group A streptococcal infections: third update on seasonal activity in England, 2024 to 2025 (https://www.gov.uk/government/publications/group-a-streptococcal-infections-seasonal-activity-in-england-2024-to-2025)

Laboratory surveillance of fungaemia due to yeasts in England: 2024 (https://www.gov.uk/government/publications/candidaemia-annual-data-from-voluntary-surveillance)

Laboratory confirmed cases of measles, mumps and rubella in England: October to December 2024 [Version 2] (https://www.gov.uk/government/publications/measles-mumps-and-rubella-lab-confirmed-cases-in-england-2024/laboratory-confirmed-cases-of-measles-mumps-and-rubella-in-england-october-to-december-2024)

Vaccine coverage reports in this issue

RSV vaccine coverage report in older adults for catch-up cohorts in England: April 2025 (https://www.gov.uk/government/publications/rsv-immunisation-for-older-adults-and-pregnant-women-vaccine-coverage-in-england)

RSV maternal vaccination coverage in England: January 2025 (https://www.gov.uk/government/publications/rsv-immunisation-for-older-adults-and-pregnant-women-vaccine-coverage-in-england)

Environmental and chemical hazards

Environmental Public Health Surveillance System (EPHSS) report for 2021 to 2023 (https://www.gov.uk/government/publications/environmental-public-health-surveillance-system)

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